THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Muddy water is a: 1

(a) suspension **(b)** solution **(c)** compound **(d)** pure substance

2. When a solute dissolves in water, the mass of the solution is: 1

(a) sum of the masses of solute and solvent **(b)** less than the solvent
(c) zero **(d)** equal to the solvent

3. A solution in which no more solute can dissolve at a given temperature is: 1

(a) saturated **(b)** suspension **(c)** unsaturated **(d)** dilute

4. A solution does not scatter light because its particles are: 1

(a) magnetic **(b)** very large **(c)** heavy **(d)** very small

5. A solution of copper sulphate is blue because it contains: 1

(a) chlorine **(b)** copper ions in water
(c) oxygen **(d)** sugar

6. A solution that can dissolve more solute at the same temperature is: 1

(a) a suspension **(b)** insoluble **(c)** unsaturated **(d)** saturated

7. Milk is a: 1

(a) solution **(b)** colloid **(c)** element **(d)** pure compound

8. Which gas is dissolved in water that aquatic animals need? 1

(a) chlorine **(b)** oxygen **(c)** hydrogen **(d)** helium

9. The maximum amount of solute that dissolves in 100 g of solvent at a given temperature is its: 1

(a) solubility **(b)** molarity **(c)** density **(d)** dilution

10. Which can be separated by filtration? 1

(a) alloy **(b)** sugar solution **(c)** salt solution **(d)** suspension

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the effect of temperature on the solubility of a solid in water? 2

12. What is a concentrated and a dilute solution? 2

13. Differentiate between a saturated and an unsaturated solution. 2

14. Why do aquatic animals survive in water? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. List three properties of a true solution. 3

16. Give three examples of solutions in which the solute or the solvent is a gas. 3

17. Describe how to separate salt from salt solution and sand from muddy water. 3

OR

List three properties of a suspension. 3

18. Name the solute and solvent in 3
(a) salt water
(b) soda water
(c) tincture of iodine.

OR

Why does the level of water remain nearly unchanged when a small amount of salt is dissolved in it? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student adds salt in 100 mL of water in small portions with stirring. Describe the steps to prepare a saturated solution and test it. 5

OR

Explain the use of solutions in medicine, cooking and cleaning with examples. Mention concentration. 5

20. Calculate the mass percentage of a solution containing 25 g of sugar in 175 g of water and find the sugar needed for 400 g of a solution of the same strength. 5

OR

Describe how you can distinguish between a saturated and an unsaturated solution experimentally. 5

END OF QUESTION PAPER — Total: 40 Marks

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